



2013

Preliminary Course

FINAL EXAMINATION

Friday, September 6

Mathematics

General Instructions

- o Reading Time – 5 minutes.
- o Working Time – 3 hours.
- o Write using a black pen.
- o Approved calculators may be used.
- o All necessary working should be shown for every question.

Total marks (100)

Section I

10 marks

- o Attempt Questions 1 – 10
- o Answer on the multiple choice answer sheet provided
- o Allow approximately 15 minutes for this section

Section II

90 marks

- o Attempt Questions 11 – 16
- o Answer in the booklets provided
- o Begin each question in a new booklet
- o Allow approximately 2 hours 45 minutes for this section

Name: _____

Student Number: _____

Teacher: _____

- P2** provides reasoning to support conclusions which are appropriate to the context.
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities.
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques.
- P5** understands the concept of a function and the relationship between a function and its graph.

Section I

10 marks

Attempt Questions 1-10

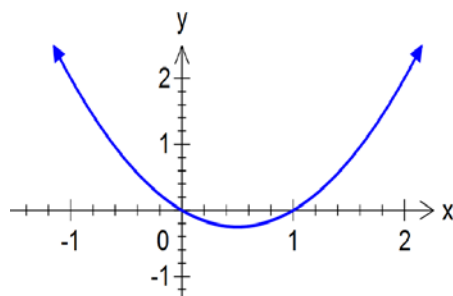
Allow about 15 minutes for this section

Use the multiple choice answer sheet for your responses to Questions 1 – 10.

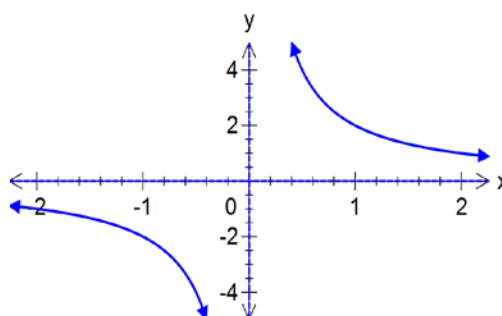
- 1 What is the value of $\frac{18.81-3.47}{2.79+7.75}$ correct to two significant figures?
- (A) 1.4
(B) 1.45
(C) 1.46
(D) 1.5
- 2 Which of the following equations has solutions $x = 2$ and $x = -3$?
- (A) $x^2 - 5x - 6 = 0$
(B) $x^2 + 5x - 6 = 0$
(C) $x^2 - x - 6 = 0$
(D) $x^2 + x - 6 = 0$
- 3 Which statement is incorrect?
- (A) The diagonals of a rhombus bisect each other
(B) The diagonals of a rhombus are equal
(C) The diagonals of a rhombus are perpendicular to each other
(D) The diagonals of a rhombus bisect the vertex angles
- 4 The function $f(x) = \frac{x^2 - 1}{x}$ is:
- (A) an even function
(B) an odd function
(C) neither an even nor odd function
(D) a zero function

5 Which graph does not represent a function?

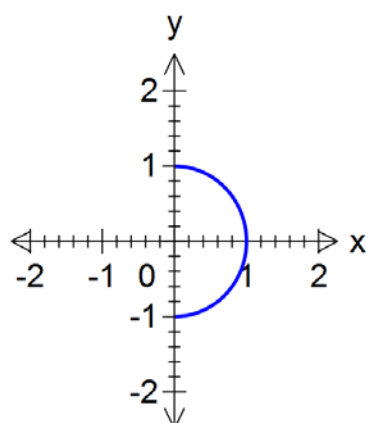
(A)



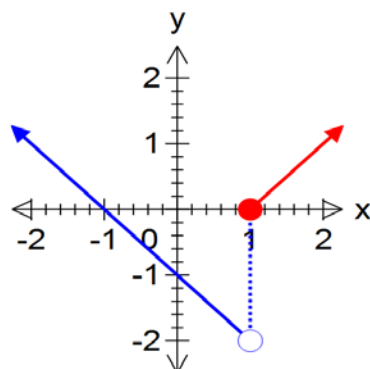
(B)



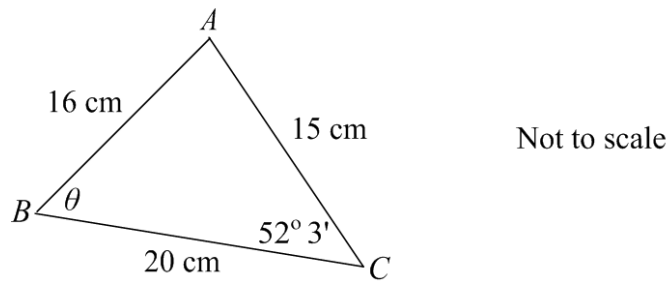
(C)



(D)



- 6 Which of the following is a correct expression involving θ in triangle ABC ?



- (A) $15^2 = 16^2 + 20^2 + 2 \times 16 \times 20 \cos \theta$
- (B) $\cos \theta = \frac{16^2 + 20^2 - 15^2}{2 \times 20 \times 15}$
- (C) $\frac{15}{\sin \theta} = \frac{16}{\sin 52^\circ 3'}$
- (D) $\frac{\sin \theta}{16} = \frac{\sin 52^\circ 3'}{15}$
- 7 What is the gradient of the line perpendicular to the line $2x + y + 3 = 0$?
- (A) -2
- (B) $-\frac{1}{2}$
- (C) $\frac{1}{2}$
- (D) 2
- 8 The solution to $t^2 > t$ is:
- (A) $0 < t < 1$
- (B) $t < 0$ or $t > 1$
- (C) $t > 1$
- (D) $t < -1$ or $t > 0$

9 What is the exact value of $\cos 240^\circ$?

(A) $\frac{1}{2}$

(B) $-\frac{1}{2}$

(C) $\frac{\sqrt{3}}{2}$

(D) $-\frac{\sqrt{3}}{2}$

10 Simplify $\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)}$

(A) 1

(B) $\cot \theta$

(C) $\tan \theta$

(D) $-\tan \theta$

End of Section I

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours 45 minutes for this section

Begin each question in a new booklet

All necessary working should be shown

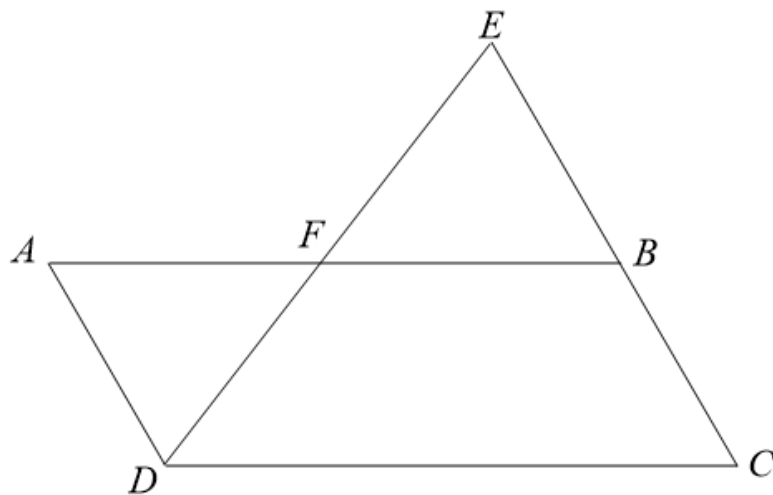
All questions are of equal value

Question 11 (15 marks).		Use a SEPARATE writing booklet.	Marks
(a)	Solve $5x - 2 = x + 10$		1
(b)	Factorise:		
(i)	$2x^2 + 7x - 4$		1
(ii)	$(x - 2)^2 - 16$		2
(c)	The line $6x - ky = 2$ passes through the point $(3, 2)$. Find the value of k .		2
(d)	Solve the following pair of simultaneous equations: $x + y = 1$ $5x - 4y = 14$		3
(e)	Find the value of x if $\sqrt{75} + \sqrt{27} = \sqrt{x}$		3

Question 11 continues on the next page

- (f) $ABCD$ is a parallelogram. CB is produced to E so that $CB = BE$.

3



Prove $\triangle AFD \equiv \triangle EFB$

Question 12 (15 marks). Use a SEPARATE writing booklet.

Marks

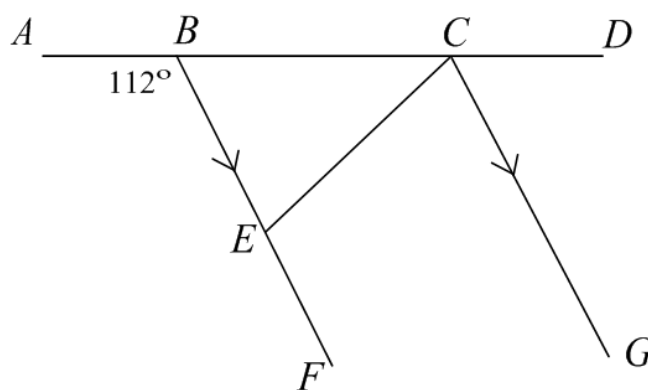
(a) Solve $\frac{3}{x-2} - \frac{5}{2} = 1$ 3

(b) Solve $2 + 3x = |x + 1|$ 3

(c) What are the values of a and b if $\frac{5-2\sqrt{2}}{1+\sqrt{2}} = a + b\sqrt{2}$? 3

(d) Prove $\tan \theta + \sec \theta = \frac{1 + \sin \theta}{\cos \theta}$ 2

(e) BF is parallel to CG , $BC = EC$ and $\angle ABE = 112^\circ$.



Not to scale

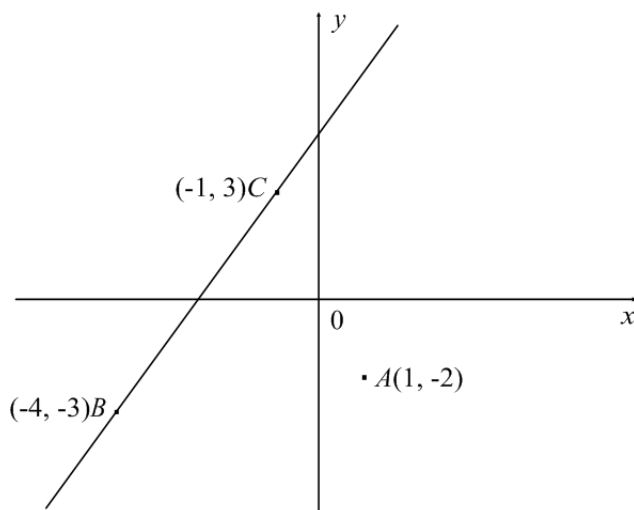
(i) Show that $\angle BEC = 68^\circ$. 2

(ii) Hence, or otherwise, show that CG bisects $\angle DCE$. 2

Question 13 (15 marks). Use a SEPARATE writing booklet.

Marks

- (a) In the diagram below the points A , B and C have coordinates $(1, -2)$, $(-4, -3)$ and $(-1, 3)$ respectively.

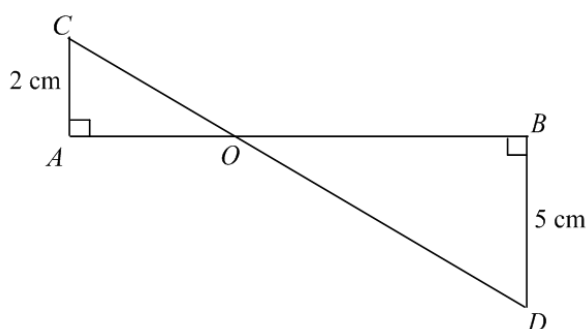


- | | | |
|-------|--|---|
| (i) | Calculate the exact length of interval BC | 2 |
| (ii) | Find the gradient of BC | 1 |
| (iii) | Hence, show that the equation of BC is $y = 2x + 5$ | 1 |
| (iv) | Find, to the nearest degree, the acute angle between the x -axis and the line BC | 2 |
| (v) | Find the perpendicular distance between A and the line BC | 2 |
| (vi) | Find the coordinates of D , in the first quadrant, so that $ABCD$ is a parallelogram | 2 |
| (vii) | Find the exact area of the parallelogram $ABCD$ | 1 |
- (b) The lengths of the sides of triangle ABC are $a = 5.2$ cm, $b = 7.3$ cm and $c = 6.7$ cm.
- | | | |
|-------|--|---|
| (i) | Explain why $\angle BAC$ is the smallest angle in the triangle. | 1 |
| (ii) | Calculate the size of the smallest angle in $\triangle ABC$. Give the answer correct to the nearest minute. | 2 |
| (iii) | Hence find the area of the triangle. Give the answer correct to the nearest square centimetre. | 1 |

Question 14 (15 marks). Use a SEPARATE writing booklet.

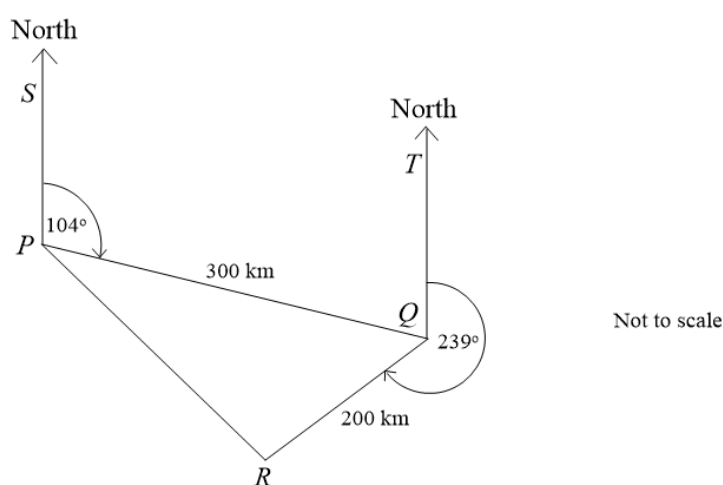
Marks

- (a) In the diagram, AOB and COD are straight lines. $AC \perp AB$ and $AB \perp BD$.



- | | | |
|------|---|---|
| (i) | Prove $\triangle ACO \parallel \triangle BDO$ | 2 |
| (ii) | If CD is 35 cm, find the length of OD | 2 |

- (b) A ship leaves port P and travels on a bearing of 104° a distance of 300 km to point Q . It then turns and travels on a bearing of 239° for 200 km to point R .



- | | | |
|-------|--|---|
| (i) | Show that $\angle PQR = 45^\circ$. | 2 |
| (ii) | What is the distance from R to P ?
Answer correct to the nearest kilometre. | 2 |
| (iii) | Find the bearing of R from P ?
Answer correct to the nearest degree. | 3 |
-
- | | | |
|-----|---|---|
| (c) | (i) Show that $2\cos^2 x - 2 + 3\sin^2 x = \sin^2 x$ | 2 |
| | (ii) Hence, or otherwise, solve $2\cos^2 x - 2 + 3\sin^2 x = 1$ for $0^\circ \leq x \leq 360^\circ$ | 2 |

Question 15 (15 marks). Use a SEPARATE writing booklet.

Marks

(a) The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} x-1 & \text{for } x \leq -2 \\ -1 & \text{for } -2 < x < 1 \\ x+1 & \text{for } x \geq 1 \end{cases}$$

(i) Evaluate $f(-2) + f(1)$ 2

(ii) Write an expression for $f(a^2 + 1)$ 1

(b) Make neat sketches of the following graphs on separate number planes. Mark clearly the essential features of each graph.

(i) $(x-3)^2 + (y+4)^2 = 25$ 2

(ii) $y = 4 - x^2$ 2

(iii) $xy = 2$ 2

(iv) $y = 1 - 2^{-x}$ 2

(c) Solve $|7 - 3x| < 3$ and graph your solution on a number line. 3

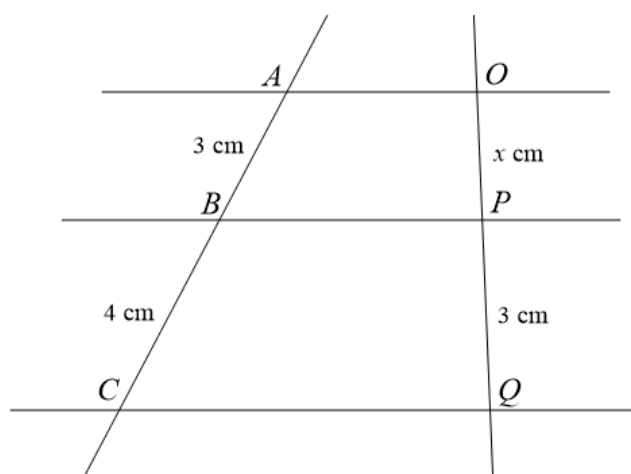
(d) If $\tan \theta = -\frac{4}{5}$ and $\cos \theta > 0$, is the value of $\sin \theta$ positive or negative? 1

Question 16 (15 marks). Use a SEPARATE writing booklet.

Marks

- (a) In the diagram below $AO \parallel BP \parallel CQ$.

2



Find the value of x

- (b) Graph the region represented by the inequalities:

2

$$x^2 + y^2 > 25 \text{ and } x \leq 0$$

- (c) Simplify $\frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{4x^2 + 4x + 4}$

3

- (d) Find the domain of $y = \frac{1}{\sqrt{5-x}}$.

2

- (e) Find the equation of the line through the point of intersection of the lines $6x - 5y = 3$ and $4x + y = -11$ and also through the point $(2, 1)$

3

- (f) If the points $(-2a, 3)$, $(a-1, a-2)$ and $(a-3, a+1)$ are collinear, find the value of a .

3

End of Examination Paper

Section I Answer Sheet

Student Number: _____

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use this multiple choice answer sheet for questions 1 – 10.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
correct

Completely fill the response oval representing the most correct answer.

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☐ D ☐

8 A ☐ B ☐ C ☐ D ☐

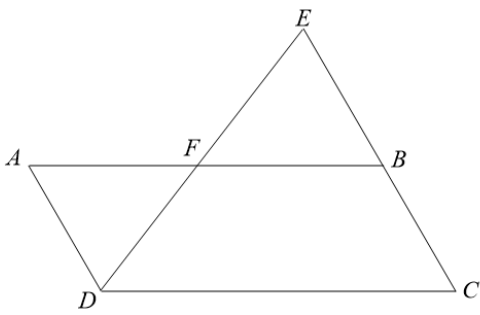
9 A ☐ B ☐ C ☐ D ☐

10 A ☐ B ☐ C ☐ D ☐

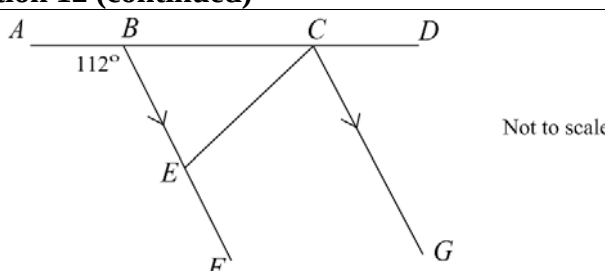
Solutions to Year 11 Mathematics Preliminary Examination 2013

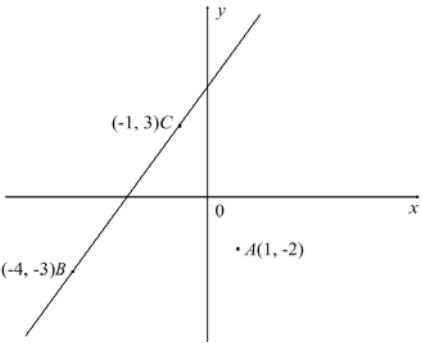
Section I Multiple Choice Solutions	
1	$\frac{18.81 - 3.47}{2.79 + 7.75}$ $= 1.455\dots$ $= 1.5 \text{ (2sf)} \quad \text{(D)}$
2	$(x - 2)(x + 3) = 0$ $x^2 + x - 6 = 0 \quad \text{(D)}$
3	The diagonals of a rhombus are equal (B)
4	$f(x) = \frac{x^2 - 1}{x}$ $f(-x) = \frac{(-x)^2 - 1}{(-x)}$ $= -\frac{x^2 - 1}{x}$ $= -f(x) \quad \text{(B)}$
5	For some x values there are two matching y values (C)
6	$\frac{15}{\sin \theta} = \frac{16}{\sin 52^\circ 3'}$ (C)
7	$2x + y + 3 = 0$ $y = -2x - 3$ $m = -2$ $m_{\perp} = \frac{1}{2} \quad \text{(C)}$
8	$t^2 > t$ $t^2 - t > 0$ $t(t - 1) > 0$ $t < 0 \text{ or } t > 1 \quad \text{(B)}$
9	$\cos 240^\circ$ $= \cos(180^\circ + 60^\circ)$ $= -\cos 60^\circ$ $= -\frac{1}{2} \quad \text{(B)}$
10	$\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)}$ $= \frac{\sin \theta}{\cos \theta}$ $= \tan \theta \quad \text{(C)}$

Question 11		Marking Criteria	
(a)	$5x - 2 = x + 10$ $4x = 12$ $x = 3$	1	Correct answer
(b) (i)	$2x^2 + 7x - 4$ $= (2x - 1)(x + 4)$	1	Correct answer
(b) (ii)	$(x - 2)^2 - 16$ $= (x - 2 + 4)(x - 2 - 4)$ $= (x + 2)(x - 6)$	2	Correct solution
		1	Attempt at difference of two squares OR Correct expansion $x^2 - 4x - 12$
(c)	Sub. (3, 2) into $6x - ky = 2$: $6(3) - k(2) = 2$ $2k = 16$ $k = 8$	2	Correct solution
		1	Correct substitution of (3, 2) OR Correct answer without justification
(d)	$x + y = 1$(1) $5x - 4y = 14$(2) $(1) \times 5 : 5x + 5y = 5$(3) $(3) - (2) : 9y = -9$ $y = -1$ In (1): $x - 1 = 1$ $x = 2$	3	Correct solution
		2	Correct elimination/substitution method with only one correct value for either x or y
		1	Correct attempt at either elimination or substitution method
e)	$\sqrt{75} + \sqrt{27}$ $= 5\sqrt{3} + 3\sqrt{3}$ $= 8\sqrt{3}$ $= \sqrt{64 \times 3}$ $= \sqrt{192}$ $\therefore x = 192$	3	Correct solution $x = 192$
		2	Substantially correct solution
		1	Correct attempt at simplifying the surds

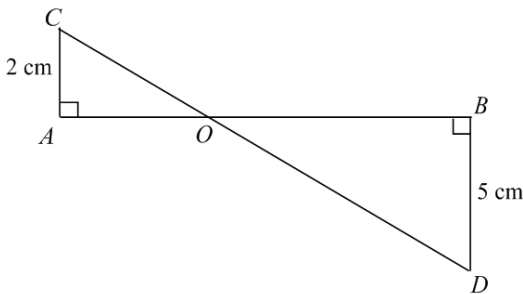
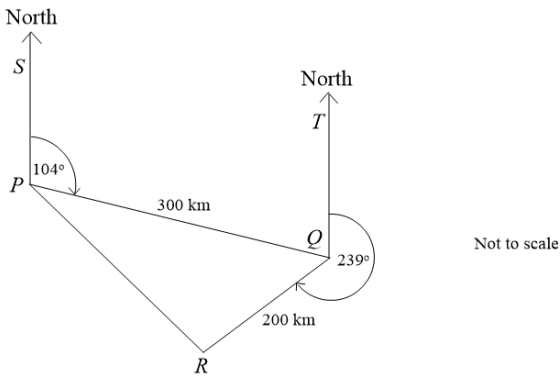
Question 11 (continued)		Marking Criteria	
(f)	 $AD = BC$ (equal opposite sides in parallelogram $ABCD$) $CB = BE$ (given) $\therefore AD = EB$ (both equal BC) $\angle AFD = \angle BFE$ (equal vertically opposite angles) $\angle DAF = \angle EBF$ (equal alternate angles, $AD \parallel EC$) $\triangle AFD \equiv \triangle EFB$ (SAA)	3	Correct solution
		2	Two correct statements, justified OR Correct proof but not fully justified
		1	One correct statement, justified OR Correct proof without any justification
Communication:			
d) clear setting out and labelling of equations 1 mark			
f) articulate reasoning, clear argument 1 mark			

Question 12		Marking Criteria	
(a)	$\frac{3}{x-2} - \frac{5}{2} = 1$ $\frac{3}{x-2} = \frac{7}{2}$ $7x - 14 = 6$ $7x = 20$ $x = \frac{20}{7}$	3	Correct solution
		2	Substantially correct solution
		1	Correct attempt at solution
(b)	$2 + 3x = x + 1 $ $2 + 3x = x + 1 \quad \text{or} \quad 2 + 3x = -x - 1$ $2x = -1 \qquad 4x = -3$ $x = -\frac{1}{2} \qquad x = -\frac{3}{4}$ Solutions are valid if: $2 + 3x \geq 0$ i.e. $x \geq -\frac{2}{3}$ $\therefore x = -\frac{1}{2}$ is only valid solution OR Check solutions by substitution into equation	3	Correct solution
		2	One mistake in the solution
		1	Correct attempt at solution
(c)	$\frac{5 - 2\sqrt{2}}{1 + \sqrt{2}} \times \frac{1 - \sqrt{2}}{1 - \sqrt{2}}$ $= \frac{5 - 5\sqrt{2} - 2\sqrt{2} + 4}{1 - 2}$ $= \frac{9 - 7\sqrt{2}}{-1}$ $= -9 + 7\sqrt{2}$ $\therefore a = -9$ and $b = 7$	3	Correct solution
		2	One mistake in working towards the answer
		1	Correctly attempting to rationalise the denominator
(d)	$\tan \theta + \sec \theta = \frac{1 + \sin \theta}{\cos \theta}$ LHS = $\tan \theta + \sec \theta$ $= \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}$ $= \frac{\sin \theta + 1}{\cos \theta}$ = RHS OR starting from RHS show it equals LHS	2	Correct solution with correct setting out
		1	Correct attempt at proof by rewriting expression in terms of $\sin \theta$ and/or $\cos \theta$ OR equivalent

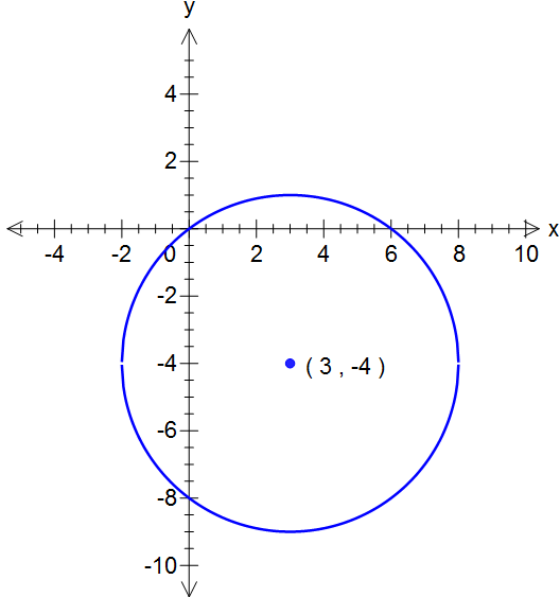
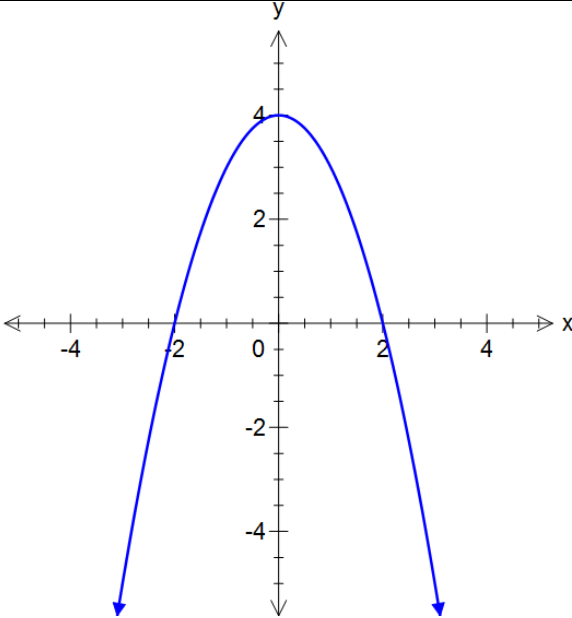
Question 12 (continued)		Marking Criteria	
(e) (i)	 Not to scale $\angle CBE = 68^\circ$ (supplement of $\angle ABE$) $\angle CBE = \angle BEC = 68^\circ$ (angles opposite equal sides in isosceles $\triangle CBE$)	2	Correct proof
		1	One correct statement, justified OR Correct proof, not fully justified
(ii)	$\angle CBE = \angle DCG = 68^\circ$ (equal corresponding angles, $BF \parallel CG$) $\angle BEC = \angle ECG = 68^\circ$ (equal alternate angles, $BF \parallel CG$) CG bisects $\angle DCE$ ($\angle ECG = \angle GCD$)	2	Correct proof
		1	One correct statement, justified OR Correct proof, not fully justified
Communication: (e) clear, concise and correct setting out of proofs 4 marks			

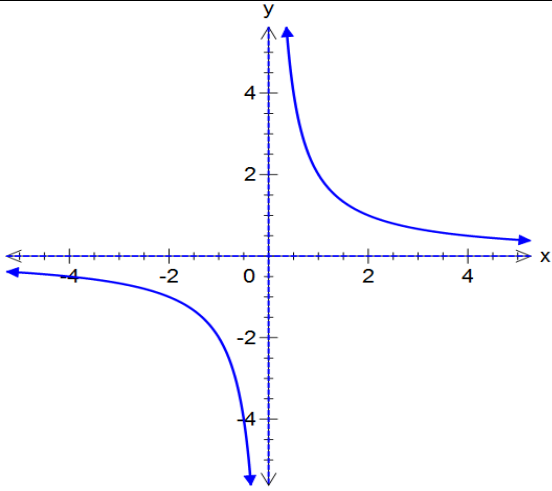
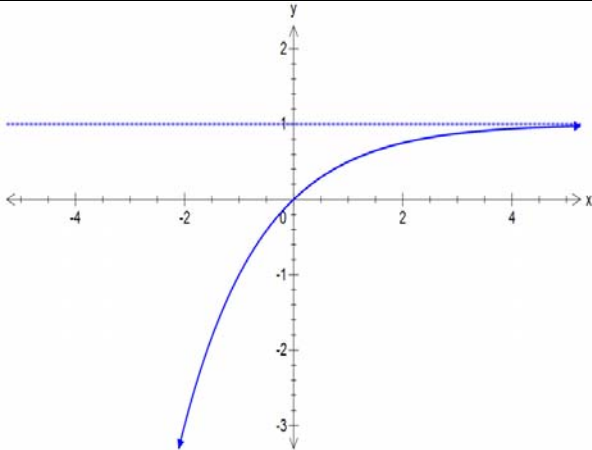
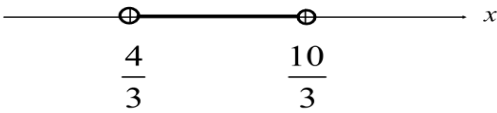
Question 13		Marking Criteria	
(a) (i)	 $BC = \sqrt{3^2 + 6^2}$ $= \sqrt{45}$ $= 3\sqrt{5} \text{ units}$	2	Correct answer
		1	Correct length as an unsimplified surd
(ii)	$m = \frac{6}{3}$ $= 2$	1	Correct answer
(iii)	$y - 3 = 2(x + 1)$ $y - 3 = 2x + 2$ $y = 2x + 5$	1	Correct solution
(iv)	$\tan \theta = 2$ $\theta = \tan^{-1}(2)$ $= 63.434\dots$ $= 63^\circ \text{ (nearest deg)}$	2	Correct solution
		1	Correct statement $\tan \theta = 2$
(v)	$d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 2(1) - 1(-2) + 5 }{\sqrt{2^2 + 1^2}}$ $= \frac{9}{\sqrt{5}} \text{ units}$	2	Correct solution
		1	Correct substitution into correct formula
(vi)	$B \rightarrow A: \text{Right } 5, \text{ up } 1$ $C \rightarrow D: \text{Right } 5, \text{ up } 1 (CD = BA, CD \parallel BA)$ $\therefore D = (-1 + 5, 3 + 1)$ $= (4, 4)$	2	Correct answer
		1	One correct coordinate

Question 13 (continued)		Marking Criteria	
(vii)	$A = bh$ $= 3\sqrt{5} \times \frac{9}{\sqrt{5}}$ $= 27 \text{ units}^2$	1	Correct answer
(b) (i)	$\angle BAC$ is opposite the shortest side	1	Correct reason
(ii)	$\cos \angle BAC = \frac{7.3^2 + 6.7^2 - 5.2^2}{2 \times 7.3 \times 6.7}$ $= 0.72725\dots$ $\angle BAC = \cos^{-1}(0.72725\dots)$ $= 43^\circ 20' 35.92''$ $= 43^\circ 21' \text{ (nearest min)}$	2	Correct solution rounded correctly
		1	Correct substitution into cosine rule OR Substantially correct solution
(iii)	$A = \frac{1}{2}bc \sin A$ $= \frac{1}{2} \times 7.3 \times 6.7 \sin 43^\circ 20' 35''$ $= 16.785\dots$ $= 17 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$ OR $A = \frac{1}{2} \times 7.3 \times 6.7 \sin 43^\circ 21'$ $= 16.787\dots$ $= 17 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$	1	Correct solution
Communication: a) v) showing substitution 1 mark vi) explaining how coordinates are found 1 mark b) ii) showing unrounded answer in degrees/minutes/seconds before rounding 1 mark			

Question 14		Marking Criteria	
(a) (i)	 $\angle A = \angle B = 90^\circ$ (given) $\angle AOC = \angle BOD$ (equal vertically opposite angles) $\triangle ACO \parallel \triangle BDO$ (equiangular)	2	Correct solution
		1	One correct statement, justified OR Correct proof, not fully justified
(ii)	$\frac{AC}{BD} = \frac{CO}{DO} = \frac{2}{5}$ (corresponding sides in similar triangles are in same ratio) $OD = \frac{5}{7} \times 35$ $= 25 \text{ cm}$	2	Correct solution, justified
		1	Correct solution, not fully justified OR Correct first statement with reason
(b) (i)	 $\angle PQT = 76^\circ$ (supplementary cointerior angles on parallel lines) $\angle TQR = 121^\circ$ (angles at point add to 360°) $\angle PQR = 45^\circ$ ($\angle TQR - \angle PQT$)	2	Correct proof
		1	Correctly finding $\angle PQT$ or $\angle TQR$, justified OR Correct proof, not fully justified
(ii)	$RP^2 = 300^2 + 200^2 - 2 \times 300 \times 200 \cos 45^\circ$ $= 45147.18\dots$ $RP = \sqrt{45147.18\dots}$ $= 212.47\dots$ $= 212 \text{ km (nearest km)}$	2	Correct solution
		1	Correct substitution in cosine rule

Question 14 (continued)		Marking Criteria	
(iii)	$\frac{\sin \angle QPR}{200} = \frac{\sin 45^\circ}{RP}$ $\sin \angle QPR = \frac{200 \sin 45^\circ}{RP}$ $= 0.665...$ $\angle QPR = 41.726...$ Bearing of R from P $= 104^\circ + 42^\circ$ $= 147^\circ$	3	Correct solution
		2	Correct value of $\angle QPR$
		1	Correct substitution into sine rule
(c) (i)	$2 \cos^2 x - 2 + 3 \sin^2 x = \sin^2 x$ LHS = $2 \cos^2 x - 2 + 3 \sin^2 x$ $= 2(1 - \sin^2 x) - 2 + 3 \sin^2 x$ $= 2 - 2 \sin^2 x - 2 + 3 \sin^2 x$ $= \sin^2 x$ $= \text{RHS}$	2	Correct proof
		1	Correct attempt at proof
(ii)	$2 \cos^2 x - 2 + 3 \sin^2 x = 1$ $\sin^2 x = 1$ $\sin x = \pm 1$ $x = 90^\circ, 270^\circ$	2	Correct solution
		1	One correct value for x
Communication: (a)(i) 1 mark for copying diagram 1 mark for stating vertices in corresponding order 1 mark for sufficient proof i.e. equiangular when two pairs of corresponding angles are equal (ii) 1 mark for correct reason why corresponding pairs of sides are in same ratio 1 mark for correct units for OD (b)(i) 2 marks for each of correct geometric reasons to show angle $PQR = 45^\circ$ (ii) 1 mark for correct rounding (iii) 1 mark for correct rounding			

Question 15		Marking Criteria	
(a) (i)	$f(-2) + f(1)$ $= (-2 - 1) + (1 + 1)$ $= -1$	2	Correct solution
		1	Correct attempt at solution
(ii)	$f(a^2 + 1) = a^2 + 1 + 1$ $= a^2 + 2$	1	Correct answer
(b) (i)		2	Correct graph showing all essential features
		1	Correct graph without all essential features shown
(ii)		2	Correct graph showing all essential features
		1	Correct graph without all essential features shown

Question 15 (continued)		Marking Criteria	
(iii)		2	Correct graph showing all essential features, including asymptotes
		1	Correct graph without all essential features shown
(iv)		2	Correct graph showing all essential features, including asymptotes
		1	Correct graph without all essential features shown
(c)	$ 7 - 3x < 3$ $7 - 3x < 3$ or $7 - 3x > -3$ $3x > 4$ $3x < 10$ $x > \frac{4}{3}$ $x < \frac{10}{3}$ $\therefore \frac{4}{3} < x < \frac{10}{3}$ 	3	Correct solution and number line graph
		2	Correct solution OR Correct attempt at solution and correct number line graph from their working
		1	Correct attempt at solution OR Correct number line graph from incorrect solution
(d)	$\tan \theta$ is negative in 2nd and 4th quadrants $\cos \theta$ is positive in 1st and 4th quadrants $\therefore \theta$ lies in 4th quadrant where $\sin \theta$ is negative	1	Correct answer
Communication: (a) Showing clear and correct substitution for (i) and (ii) 3 marks (d) Clear setting out of reason for $\sin \theta$ being negative 1 mark			

Question 16		Marking Criteria	
(a)	$\frac{x}{3} = \frac{3}{4}$ (intercepts on transversals are cut in same ratio by a family of parallel lines) $x = \frac{9}{4}$ cm	2	Correct proof with full and correct reason
		1	Correct first statement, justified OR Correct proof, not fully justified
(b)		2	Correct region
		1	One correct region
(c)	$\frac{x^3 - 1}{x^2 - 1} \times \frac{x^2 - 4x - 5}{4x^2 + 4x + 4}$ $= \frac{(x-1)(x^2 + x + 1)}{(x+1)(x-1)} \times \frac{(x-5)(x+1)}{4(x^2 + x + 1)}$ $= \frac{x-5}{4}$	3	Correct solution
		2	Substantially correct solution
		1	Correct attempt at factorisation

Question 16 (continued)		Marking Guidelines	
(d)	$y = \frac{1}{\sqrt{5-x}}, x \neq 5$ Domain: $5-x > 0$ $x < 5$	2	Correct answer
		1	Correct attempt at finding domain by stating $x \neq 5$ or $x \leq 5$
(e)	$6x-5y-3+k(4x+y+11)=0$ Sub. (2, 1): $12-5-3+k(8+1+11)=0$ $4+20k=0$ $k = \frac{-4}{20}$ $k = -\frac{1}{5}$ $6x-5y-3-\frac{1}{5}(4x+y+11)=0$ $30x-25y-15-4x-y-11=0$ $26x-26y-26=0$ $x-y-1=0$	3	Correct solution
		2	Correct value of k OR Correct point of intersection and attempt to find equation of required line
		1	Correct form of equation shown by first line and attempt to substitute (2, 1) OR Correct attempt to solve equations simultaneously
(f)	$\frac{(a-2)-3}{(a-1)+2a} = \frac{(a+1)-3}{(a-3)+2a}$ $\frac{a-5}{3a-1} = \frac{a-2}{3a-3}$ $(3a-3)(a-5) = (3a-1)(a-2)$ $3a^2-18a+15 = 3a^2-7a+2$ $11a = 13$ $a = \frac{13}{11}$	3	Correct solution
		2	Substantially correct solution
		1	Correct attempt at solution shown by first line of working

Communication: Question 16 (4 marks)

- (b) The graph is drawn neatly with template. Intercepts shown and graph labelled. **1 Mark**
- (d) Clear setting out. For example stating limitations such as $x \neq 5$. **1 Mark**
- (e) Clear and logical setting out of solution (example: mention of substitution). **1 Mark**
- (f) Clear explanation that equal gradients will prove the points are collinear. **1 Mark**